

Problem-Solution Fit

Maximum Marks: 3 Marks

Date

15 March 2026

Team ID

Batch 2023-27 | ACOE

Project Name

Rising Waters – ML-Based Flood Prediction System

Team Members

S.No	Team Member Name	Role
1	Venkatesh Balireddy	Team Lead
2	Archana Dhanani	Member
3	Shivatmika Gandikota	Member
4	Gode Siva Ramakrishna Durgaprasad	Member
5	Bavisetty Gopi Krishna	Member

Problem-Solution Fit Analysis

This document maps each identified problem to the specific solution feature in Rising Waters that addresses it, demonstrating a strong problem-solution fit.

S.No	Problem Identified	Solution Feature	Fit Level
1	No real-time ML-powered flood prediction tool available	XGBoost classifier (floods.save) achieving 96.55% accuracy	Strong
2	Complex meteorological data difficult to interpret manually	6-field standardized input form with field descriptions and tooltips	Strong
3	No emergency response guidance for prediction outputs	Flood Result page with 6 colour-coded emergency protocol action cards	Strong
4	No accessible web interface for non-technical disaster management users	Flask web app with intuitive dark-theme UI accessible on any browser	Strong
5	No prediction history or audit trail	predictions_log.csv + History page with full timestamped audit log	Strong
6	Lack of 24/7 cloud accessibility for disaster teams	IBM Cloud Foundry deployment with Docker containerization	Strong
7	Manual comparison of multiple ML models is time-consuming	Home Dashboard with built-in accuracy comparison (DT/RF/KNN/XGB)	Medium